Software Requirements Specification

for

Scholarship Portal

Version 2.0

Prepared by

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REVISIONS

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Vikrant Singh, Shubham Sinha	Baseline version of the Skill Scholar SRS created, covering core requirements and system overview.	15/08/2025
2.0	Vikrant Singh, Shubham Sinha	Revised and updated version of the Skill Scholar SRS with expanded requirements, diagrams, and detailed sections.	14/09/2025

1 INTRODUCTION

1.1 Document Purpose

This This document specifies the software requirements for the Skill Scholar platform, a web-based scholarship discovery and application system. It defines the functional, non-functional, and interface requirements to provide a clear understanding of the system's objectives, features, and constraints. The SRS serves as a contract between stakeholders, developers, and evaluators, ensuring that all parties share a common vision of the product.

The primary purpose of this document is to guide the development, testing, and future enhancement of the Skill Scholar system. Version 2.0 of the SRS expands upon the baseline (Version 1.0) by including detailed requirements, updated diagrams, and refined use cases, reflecting the current deployed version of the platform at SkillScholar.

This SRS focuses on the core modules of the system—student registration, profile creation, scholarship search, personalized recommendations, application submission, and administrator management of scholarships. Future enhancements such as mobile applications, advanced blockchain integration, and partnerships with educational institutions are not part of this version but may be addressed in subsequent revisions.

1.2 Product Scope

Skill Scholar is a web-based platform developed to streamline the process of discovering and applying for scholarships. It provides a centralized system where students can register, create profiles, search scholarships, receive personalized recommendations, and track applications. Administrators are able to manage and verify scholarship listings, ensuring accuracy and reliability.

The primary goals of the system are to:

- Improve accessibility to verified and updated scholarship opportunities.
- Reduce the complexity of application processes through guided workflows.

- Provide personalized recommendations to students based on their academic and financial profiles.
- Enhance transparency in scholarship management.

Version 2.0 of the SRS defines the functional and non-functional requirements of the core system. Future enhancements, such as mobile integration, blockchain-based verification, and international expansion, are beyond the scope of this version and may be considered in later revisions.

1.3 Intended Audience and Document Overview

This document is intended for the following stakeholders:

Project Evaluators/Faculty: To assess the completeness, clarity, and feasibility of the system requirements.
Developers: To use the requirements as a basis for system design, implementation, and testing.
Testers: To design test cases that validate the functional and non-functional requirements.
Administrators/End Users: To understand the capabilities and limitations of the system.

The document is organized as follows:

Section 1 introduces the document purpose, product scope, intended audience, definitions, conventions, and references.
Section 2 provides an overall description, including product perspective, functionality, constraints, and assumptions.
Section 3 specifies detailed functional requirements, external interface requirements, and use case models.
Section 4 outlines non-functional requirements, including performance, security, and quality attributes.
Section 5 states other requirements not covered in earlier sections.

Appendices contain supporting material such as data dictionary and group log.

Readers are advised to begin with Section 1 for an overview, then proceed to Section 2 and Section 3 for core system requirements. Sections 4 and 5 provide supporting details, while the appendices offer reference information.

1.4 Definitions, Acronyms and Abbreviations

The following terms and abbreviations are used throughout this document:

- **Admin** – System administrator responsible for managing scholarships and verifying applications.
- **AI** – Artificial Intelligence, used for generating personalized scholarship recommendations.
- **API** – Application Programming Interface, enables communication between frontend and backend services.
- **CRUD –** Create, Read, Update, Delete operations on system data.
- **DBMS** – Database Management System used to store and manage user and scholarship information.
- **IEEE –** Institute of Electrical and Electronics Engineers, provider of SRS standards.
- **ML** – Machine Learning, applied in matching algorithms for scholarship recommendations.
- **RBAC** – Role-Based Access Control for differentiating user permissions.
- **SRS** – Software Requirements Specification, this document.
- **UI** – User Interface through which students and administrators interact with the system.
- **UX** – User Experience, quality of interaction between the system and its users.

1.5 Document Conventions

The following conventions are applied throughout this document to maintain clarity, consistency, and readability:

Font and Size:

All text is in Times New Roman.

Headings/Subheadings (e.g., Document Conventions, 1.1, 1.2) are 16 pt, bold, and aligned justify.

Normal text is 12 pt, with 1.5 line spacing, aligned justify.

Text Styling:

Bold: For section headings, key terms, and important concepts.

Italic: For emphasis and examples.

Monospace / Code font: For code snippets, database table names, variables, or commands.

Numbering:

Sections and subsections follow hierarchical numbering (e.g., 1, 1.1, 1.2, 2, 2.1).

Bulleted and numbered lists are used for clarity where required.

Tables and Figures:

Tables and figures are numbered sequentially (e.g., Table 2.1, Figure 3.1) and include descriptive captions.

Cross-References:

References to other sections, tables, or figures use full numbering (e.g., see section 3.2 or Table 4.1).

1.6 References and Acknowledgments

References:

The following sources were referred to while preparing this SRS for the Skill Scholar project:

Skill Scholar official website – https://skillscholar.netlify.app

Official documentation for React.js and React Native (for frontend development).

Official documentation for Node.js and Express.js (for backend APIs).

Firebase Authentication and Realtime Database documentation.

MongoDB Atlas official guides for cloud database management.

Tutorials and guides for 3D dashboards using Three.js and GSAP.

Acknowledgments:

The authors express sincere gratitude to:

Vikrant Singh and **Shubham Sinha**, for preparing and authoring this SRS.

Skill Scholar development team, for providing insights into platform features and workflows.

Online documentation and tutorials for React.js, React Native, Node.js, Express.js, Firebase, and MongoDB Atlas, which were instrumental in understanding system functionalities.

All students and beta users who provided feedback during Skill Scholar testing.

2 OVERALL DESCRIPTION

2.1 Product Overview

Skill Scholar is a new, self-contained web-based platform designed to simplify the process of finding and applying for scholarships. Unlike existing fragmented solutions where students must manually search multiple portals, *Skill Scholar* consolidates opportunities into one system, making it easier for students to access scholarships relevant to their academic achievements, skills, and financial background.

The product functions as a standalone system but could be integrated with institutional databases in future versions. It provides separate modules for students and administrators. Students interact with the system through a user-friendly web interface, while administrators manage scholarships, verify eligibility, and update listings.

2.2 Product Functionality

The *Skill Scholar* system will provide the following major functions:

- **User Registration and Login** – Students can create accounts and log in securely.
- **Profile Management** – Students can enter academic, personal, and financial information.
- **Scholarship Search and Filtering** – Students can search scholarships based on eligibility criteria.

- **Scholarship Recommendation** – The system suggests scholarships that best fit a student's profile.
- **Application Submission** – Students can apply to scholarships directly through the platform.
- **Application Tracking** – Students can view the status of their scholarship applications.
- **Admin Module** – Administrators can add, update, and delete scholarships, and verify student applications.
- **Notification System** – (Optional/Extended) Students receive updates regarding scholarship deadlines and application results.

2.3 Design and Implementation Constraints

The development and design of *Skill Scholar* are subject to the following constraints:

- **Hardware Limitations**:
    - The system must be deployable on standard web servers with at least 8GB RAM, 4-core CPU, and 50GB storage.
- **Software/Technology Stack**:
    - **Frontend**: HTML, CSS, JavaScript
    - **Backend**: PHP or Node.js (fixed requirement for web compatibility)
    - **Database**: MySQL or PostgreSQL
- **Design Standards**:
    - The system must follow the **COMET Method** for software design ([Object-Oriented Analysis and Design Using the COMET Method, Addison-Wesley, 1999]).
    - The system must use **UML (Unified Modeling Language)** for diagrams and modeling (Object Management Group, UML Specifications).
- **Security Considerations**:
    - All passwords must be stored in encrypted form (e.g., SHA-256 or bcrypt).

   - The system must enforce role-based access (Student vs. Admin).

- **Browser Support**:
   - Must run on Chrome, Firefox, and Microsoft Edge.

2.4 Assumptions and Dependencies

The design and requirements of *Skill Scholar* are based on the following assumptions and dependencies:

- Students will have consistent access to the internet and a web browser.
- Scholarship data entered into the system is assumed to be accurate and up-to-date.
- Administrators will be responsible for verifying scholarship authenticity and managing the database.
- The system will depend on open-source frameworks and libraries (e.g., Bootstrap, jQuery, etc.).
- Future scalability depends on cloud hosting services if the number of users grows significantly.
- The success of recommendation algorithms assumes that students provide correct and complete profile data.

3 SPECIFIC REQUIREMENTS

3.1 External Interface Requirements

3.1.1 User Interfaces

UI Overview (logical characteristics & interaction):

Skill Scholar is a responsive web application accessed from desktop and mobile browsers. Primary UI patterns:

- **Login / Register Page**
  - Fields: email, password, confirm password (register: name, phone, college, stream).
  - Actions: “Login”, “Register”, “Forgot Password”.
  - Interaction: form input, client-side validation, captcha optional.
- **Student Dashboard**
  - Sections: Profile summary, Recommended Scholarships (cards), Search bar + filters, Applied Scholarships (status), Notifications/Alerts.
  - Interaction: click-to-expand scholarship cards, “Apply” button opens application form modal/page, inline edit for profile.
- **Scholarship Details Page**
  - Fields: Eligibility, Deadline, Amount, Provider info, Required Documents, Apply button.
  - Interaction: view full description, attach documents (upload), save to shortlist.
- **Admin Dashboard**
  - Sections: Manage Scholarships (list + CRUD buttons), Verify Applications, User Management, Reports.

- Interaction: table actions (edit, delete, publish/unpublish), bulk import (CSV), verification workflow (approve/reject).

- **Forms & Navigation**
  - Navigation bar with Home, Search, Dashboard, Help, Logout.
  - All forms use client-side validation and clear inline error messages.
  - Accessible labels and keyboard navigation supported.

3.1.2 Hardware Interfaces

Skill Scholar is primarily a web software product and has few hardware dependencies. Logical/physical interfaces:

- **Client Devices**
  - Desktop/laptop, tablet, smartphone (modern browsers).
  - Input: keyboard, mouse, touch.
  - Output: display (HTML), notifications (browser push).
- **Server / Hosting**
  - Web server (Linux VM / cloud instance) hosting backend and app.
  - Database server (MySQL/Postgres) hosted on same or separate instance.
- **Optional External Hardware**
  - SMS Gateway hardware/service (for OTP) — system calls gateway API.
  - SMTP mail server (cloud/email provider) — used to send verification and notifications.

Each sensor-like interaction (e.g., reading upload file size) is handled by standard HTTP multipart/form-data; all numeric units (if any) are standard web units (bytes for storage, seconds for timeouts).

3.1.3 Software Interfaces

Direct software connections / APIs:

- **Backend REST API**
  - Endpoints for authentication (/api/auth), student profile (/api/students), scholarships (/api/scholarships), applications (/api/applications), admin actions.
  - JSON request/response; Bearer token (JWT) authentication.
- **Database (SQL)**
  - Stores Students, Scholarships, Applications, Admins, Logs.
- **Email Service**
  - SMTP or transactional email provider (SendGrid/Mailgun) to send verification, password reset, notifications.
- **SMS/OTP Service (optional)**
  - External API to send OTP for phone verification.
- **Optional Mobile App**
  - Mobile app (future) will consume the same REST API and send push notifications.

3.2 Functional Requirement

Below are detailed functional requirements with identifiers.

- **F1 — User Registration & Authentication**
  - F1.1 The system shall allow new students to register with email, password, and basic profile fields.
  - F1.2 The system shall support secure login with email and password.
  - F1.3 The system shall support password reset via email.
- **F2 — Student Profile Management**

- F2.1 The system shall allow students to create and edit profiles (name, college, stream, GPA, skills, family income).
- F2.2 The system shall allow uploading required documents (PDF, images) up to configured size.

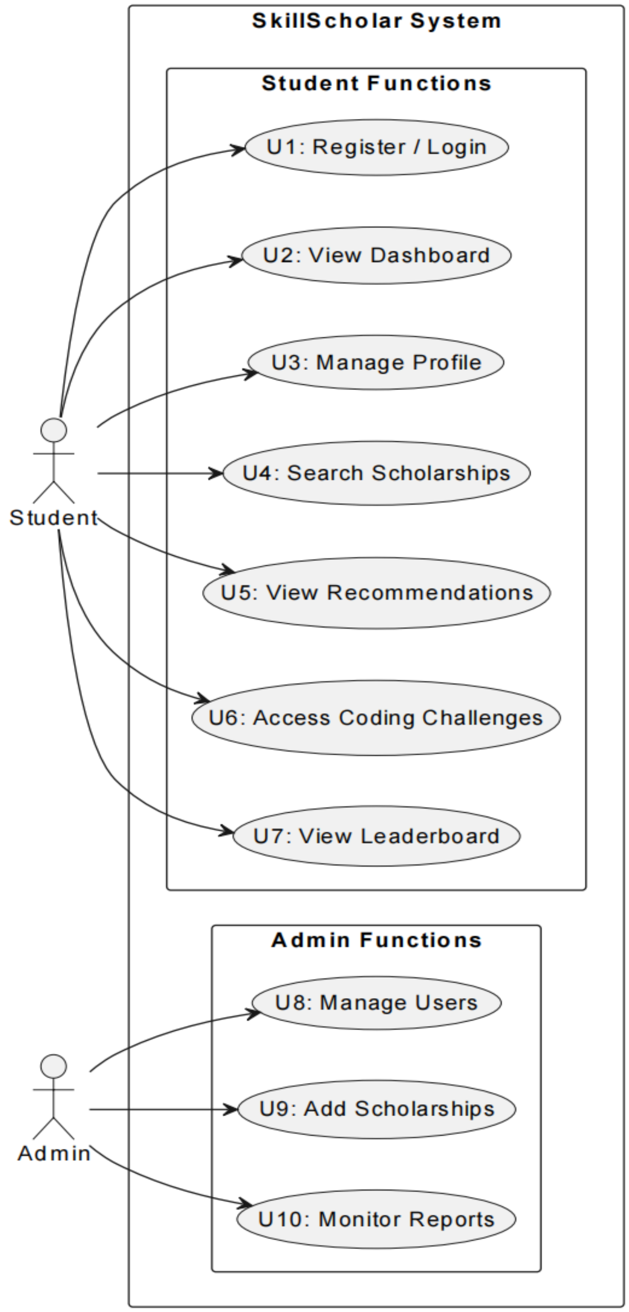
- **F3 — Scholarship Catalog & Management**
    - F3.1 The system shall allow admins to add, update, publish/unpublish, and delete scholarship records (CRUD).
    - F3.2 The system shall store scholarship metadata: title, provider, eligibility criteria, deadline, amount, required-documents, tags.
- **F4 — Search & Filter Scholarships**
    - F4.1 The system shall provide search by keywords.
    - F4.2 The system shall provide filters: eligibility (GPA, stream), deadline range, amount, provider, location, tags.
- **F5 — Recommendation Engine**
    - F5.1 The system shall generate recommended scholarships for a student based on profile matching (skills, GPA, tags).
    - F5.2 The system shall rank recommendations by match score and deadline proximity.
- **F6 — Application Workflow**
    - F6.1 The system shall allow students to apply for a scholarship by submitting required documents and form fields.
    - F6.2 The system shall create an application record with status values: Submitted, Under Review, Approved, Rejected, Withdrawn.
    - F6.3 The system shall allow students to view application history and status.
- **F7 — Admin Verification**

    - F7.1 The system shall allow admins to review applications, add notes, and change status.
    - F7.2 The system shall log admin actions for audit.
- **F8 — Notifications**
    - F8.1 The system shall send email notifications for: registration, password reset, application submission, admin decision.
    - F8.2 (Optional) The system shall support SMS/Push notifications for deadlines and decisions.
- **F9 — Security & Access Control**
    - F9.1 The system shall implement role-based access control (Student, Admin).
    - F9.2 The system shall encrypt passwords using a secure hashing algorithm (bcrypt recommended).
- **F10 — Reporting**
    - F10.1 The system shall provide admin reports: number of applications per scholarship, pending verifications, user statistics.

Each FR should later be traced to test cases in your Test Plan.

3.3 Use Case Model

User Case Diagram:



Use Case Specification – U1: Register/Login

- **Author:** Vikrant Singh
- **Purpose:** Allow a student to create an account or log into the SkillScholar system securely.
- **Requirements Traceability:** FR-1.1, FR-1.2 (User Authentication)
- **Priority:** High
- **Preconditions:**
  - User must have internet access.
  - Registration requires a unique email ID.
- **Postconditions:**
  - Student is successfully logged into the dashboard.
- **Actors:** Student
- **Extends/Includes:** None
- **Flow of Events:**
  - **Basic Flow:**
    1. Student opens SkillScholar.
    2. Student enters credentials (email, password) or clicks "Register."
    3. System validates credentials or creates a new account.
    4. Student is redirected to their dashboard.
  - **Alternative Flow:**
    - If student chooses "Forgot Password," the system sends a password reset link.
  - **Exceptions:**
    - Invalid credentials → error message displayed.
    - Already registered email → prompt user to login.
- **Notes/Issues:** Ensure password is encrypted and login process is secure (SSL).

Use Case Specification – U8: Manage Users

- **Author:** Shubham Sinha
- **Purpose:** Allow admin to view, add, block, or delete student accounts for system management.
- **Requirements Traceability:** FR-2.1 (Admin Access Control)
- **Priority:** High
- **Preconditions:**
    - Admin must be logged into the system with admin privileges.
- **Postconditions:**
    - User database is updated (additions, modifications, deletions).
- **Actors:** Admin
- **Extends/Includes:** May include “Monitor Reports” (U10).
- **Flow of Events:**
    - **Basic Flow:**
        1. Admin logs into SkillScholar.
        2. Admin opens the “Manage Users” section.
        3. Admin selects option to add, block, or delete a user.
        4. System updates user records accordingly.
    - **Alternative Flow:**
        - Admin attempts to delete a user with active session → system shows warning.
    - **Exceptions:**
        - Database connection failure.
        - Invalid admin action (e.g., trying to block self).
- **Notes/Issues:** Ensure only authorized admins can access this function; log all changes for auditing.

4 OTHER NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

- **P1:** The system shall return scholarship search results within **3 seconds** for up to **1,000 concurrent users**.
- **P2:** The system shall support at least **5,000 registered users** in the initial deployment without performance degradation.
- **P3:** The system shall handle simultaneous requests from both students and administrators with a response time not exceeding **2 seconds** for profile updates and scholarship postings.
- **P4:** The system shall ensure a **99.5% uptime** during operational hours (24/7 availability).
- **P5:** The system shall load the homepage and login page within **2 seconds** under normal broadband conditions (5 Mbps or higher).

4.2 Safety and Security Requirements

- **S1:** The system shall use **encrypted communication (HTTPS/SSL)** to protect sensitive user data during transmission.
- **S2:** User passwords shall be stored in the database using **hashing algorithms (e.g., bcrypt/argon2)**.
- **S3:** The system shall implement **role-based access control (RBAC)** to ensure that administrators, students, and scholarship providers have different access levels.
- **S4:** The system shall lock an account for **5 minutes after 3 consecutive failed login attempts** to prevent brute-force attacks.
- **S5:** The system shall comply with **data privacy regulations (e.g., GDPR principles)** ensuring that student information is not shared with third parties without consent.

- **S6:** Regular **data backups** shall be performed daily to prevent loss of information in case of system failure.

4.3 Software Quality Attributes

4.3.1 Reliability

The system shall achieve **99.5% uptime** to ensure consistent availability for students searching for scholarships. Automated error logging and recovery mechanisms will be implemented to minimize downtime and detect issues quickly.

4.3.2 Usability

The platform shall provide an **intuitive interface** with simple navigation menus and filters for scholarship search. Features like autosuggestions and guided application steps will enhance ease of use. A **help section and FAQs** will be provided for new users.

4.3.3 Maintainability

The system shall follow a **modular architecture** to allow easy updates, bug fixes, and future feature enhancements (e.g., adding AI-based recommendation engines). Documentation and code comments will be maintained to support developers during future upgrades.

4.3.4 Security

In addition to login safeguards, the system shall implement **session timeouts (15 minutes of inactivity)** and protect against common vulnerabilities such as **SQL injection, cross-site scripting (XSS), and CSRF attacks**.

4.3.5 Scalability

The system shall be designed to handle a growing user base, with database queries optimized for large-scale searches. Future versions may integrate cloud services for load balancing and higher user traffic.

5 OTHER REQUIREMENTS

The following requirements are not covered in earlier sections but are essential for the smooth functioning, compliance, and long-term sustainability of the Skill Scholar system:

5.1 Legal and Regulatory Requirements

The system shall comply with data protection and privacy regulations, including GDPR principles and relevant national IT Acts.

Scholarship data provided on the platform shall respect copyright and intellectual property laws, ensuring no unauthorized use of third-party content.

User data shall not be sold or shared with external entities without explicit consent from the student.

5.2 Ethical Requirements

The recommendation engine shall provide fair and unbiased scholarship suggestions, avoiding discrimination based on gender, caste, religion, or nationality.

User data shall only be used for improving recommendations and scholarship management.

The system shall include transparent policies regarding data collection, storage, and usage.

5.3 Environmental Requirements

The platform shall be hosted on cloud servers optimized for energy efficiency.

The system shall minimize unnecessary data transfers to reduce bandwidth usage and energy consumption.

Documentation shall be maintained in digital format to reduce paper-based records.

5.4 Support and Maintenance Requirements

A dedicated support mechanism (FAQs, chat support, or email) shall be available for students and administrators.

The system shall be maintained with regular updates, bug fixes, and security patches.

A maintenance log shall be preserved to track all updates, modifications.

5.5 Future Enhancements

Mobile application development for Android and iOS platforms.

AI-driven advanced recommendation models using deep learning.

Integration with blockchain for secure verification of scholarship authenticity.

Partnership APIs with universities and funding organizations for real-time scholarship updates.

6 APPENDIX A – DATA DICTIONARY

The following table provides a structured description of key data elements used in the Skill Scholar system:

Data Element	Description	Type	Constraints / Example
User_ID	Unique identifier for each registered user (student/faculty).	Integer / String	Auto-generated, Primary Key
Name	Full name of the user.	String	Max length: 100 characters
Email	User's registered email address, used for login and notifications.	String	Must be unique, Valid email format
Password	Encrypted password for authentication.	String (hashed)	Minimum 8 characters, stored securely
Role	Specifies user role: Student / Faculty / Admin.	Enum	Predefined values
Scholarship_ID	Unique identifier for each scholarship entry.	Integer / String	Auto-generated, Primary Key
Scholarship_Name	Official name/title of the scholarship.	String	Max length: 150 characters
Eligibility	Criteria defining scholarship eligibility.	Text	Example: "CGPA ≥ 7.0, Annual Income ≤ 3L"
Deadline	Application deadline for scholarship.	Date	Format: DD-MM-YYYY
Reward	Scholarship amount or benefits.	String / Numeric	Example: "₹50,000 per year"
Recommendation_Score	AI-generated ranking score for personalized scholarship suggestion.	Float	Range: 0.0 – 1.0
Feedback_ID	Unique identifier for user	Integer /	Auto-generated

Data Element	Description	Type	Constraints / Example
	feedback entries.	String	
Feedback_Text	User-provided comments or reviews.	Text	Max 500 characters
Timestamp	System-generated date and time of record creation or modification.	DateTime	Format: DD-MM-YYYY HH:MM:SS

7 APPENDIX B – GROUP LOG

The following table records the contributions of group members during the preparation of the Skill Scholar SRS (Version 2.0).

Member Name	Role / Responsibility	Contributions	Date(s)
Vikrant Singh	Primary Author, Project Lead	Drafted core sections of the SRS (1.0 to 4.3), refined functional requirements, coordinated documentation.	01/08/2025 – 14/09/2025
Shubham Sinha	Co-Author, Technical Contributor	Expanded requirements in Version 2.0, added detailed use case models, reviewed diagrams and data dictionary.	01/09/2025 – 14/09/2025
Both Members	Review and Finalization	Proofreading, formatting, cross-checking references, ensuring consistency and completeness of the document.	14/09/2025